

RAJIV GANDHI COLLEGE OF ENGINEERING & TECHNOLOGY, PUDUCHERRY.
DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

QUESTION BANK

BASIC ELECTRICAL & ELECTRONICS ENGINEERING

ELECTRICAL

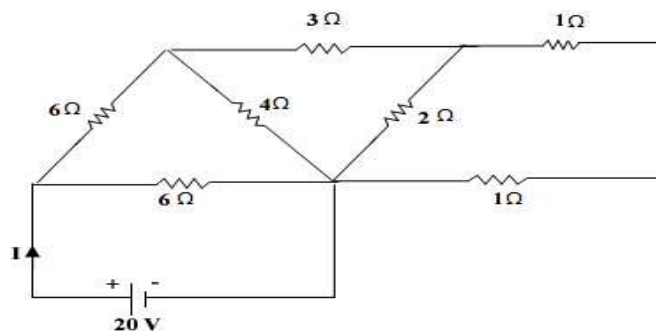
Unit-I DC CIRCUITS

PART-A

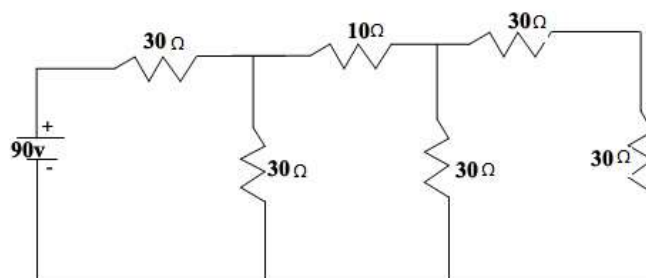
1. State Ohm's law.
2. Mention two difference between series and parallel circuits.
3. What are the limitations of Ohm's law?
4. State Kirchoff's voltage law?
5. State Kirchoff's current law?
6. Define current and voltage with their units?
7. What is current division rule?
8. What is voltage division rule?
9. What is active element?
10. What is passive element?
11. Difference between active and passive element?
12. Two resistors 4Ω and 6Ω are connected in parallel. If the total current is 12A, find the current through each resistor.

PART-B:

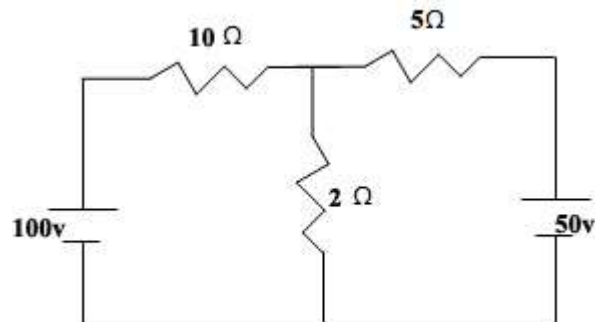
1. Find the current supplied by the source in the circuit shown in the figure below. Also determine the power in each resistance.



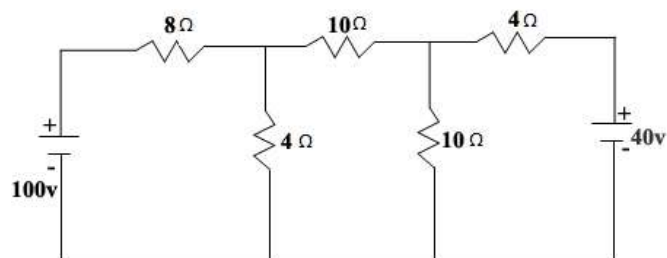
2. In the circuit given obtain the load current and power delivered to the load.



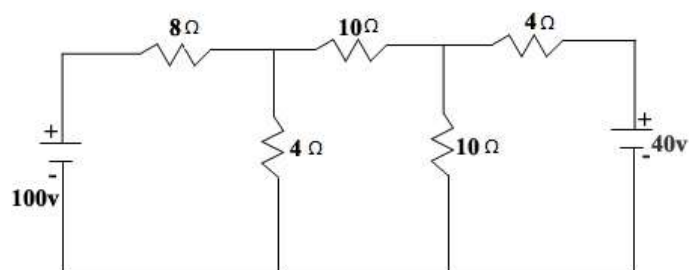
3. Using Nodal analysis, determine the current in the 2Ω resistor?



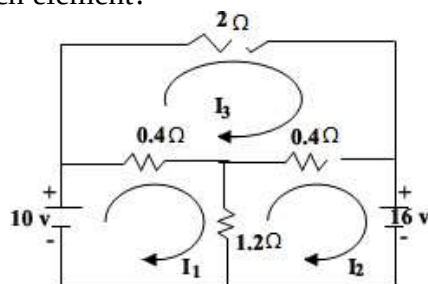
4. Derive the expression for the equivalent resistance of three resistors connected in parallel.
 5. Derive the expression for the delta to star transformations of three resistors.
 6. Derive the expression for the equivalent resistance of three resistors connected in series.
 7. Derive the expression for the star to delta transformations of three resistors.
 8. Write the mesh equations of the network by inspection and find the power absorbed by 8Ω .



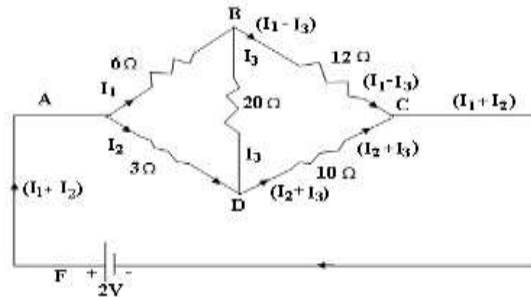
9. Using nodal analysis determine the current in the 10Ω resistor.



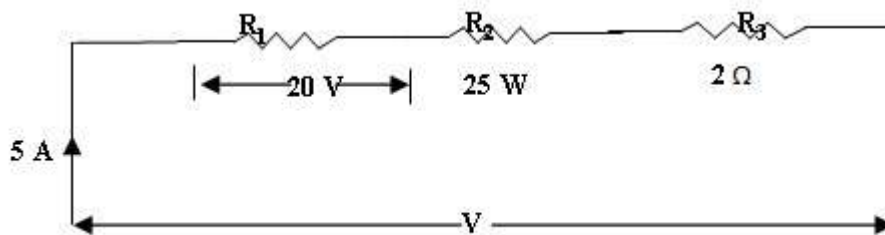
10. Find the current through each element?



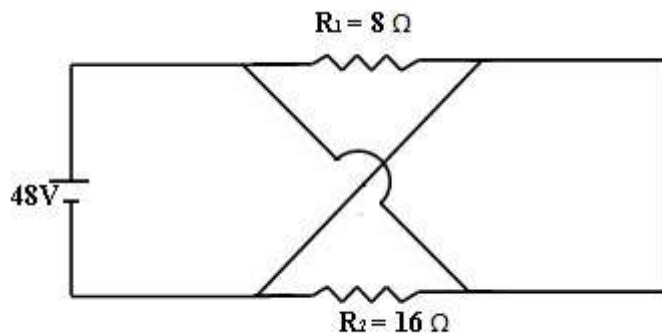
11. In the bridge network $AB = 6\Omega$; $BC = 12\Omega$; $CD = 10\Omega$; $DA = 3\Omega$; $BD = 20\Omega$ and as an emf. of $2V$. Calculate the I delivered to the battery?



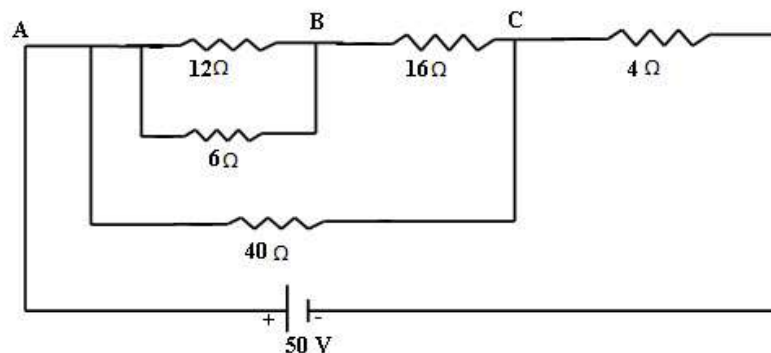
12. Three resistors R_1 , R_2 and R_3 are connected in series across a constant voltage of “V” volts. The voltage drop across R_1 is 20 Volts. The power loss in R_2 is 25 Watts and R_3 has a resistance of 2Ω . Find the voltage “V” if the current is 5 Amps.



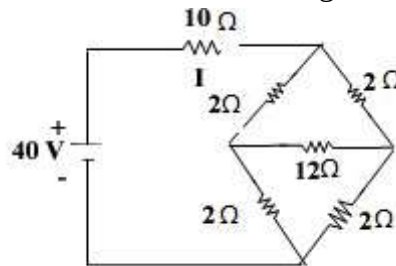
13. Calculate the current supplied by the battery in the given circuit shown below



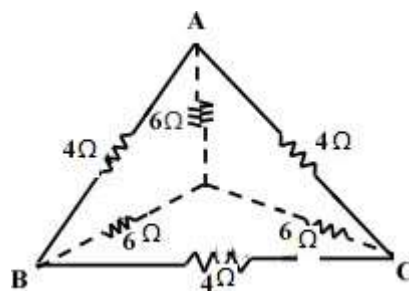
14. Calculate the equivalent resistance of the following combination of resistors and also the source current



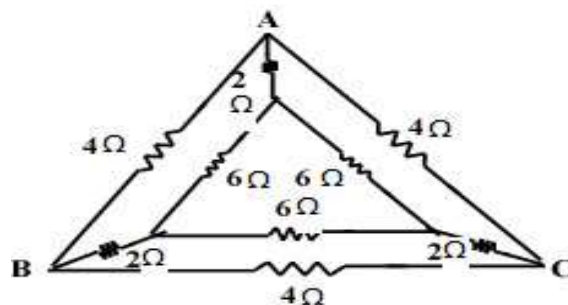
15. For the circuit shown below, find the current flowing through the $10\ \Omega$ resistor.



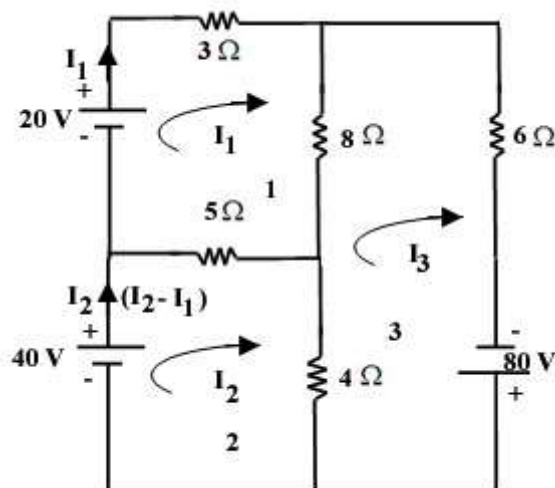
16. Calculate the equal resistance across AB of the circuit by using Star and Delta transformation



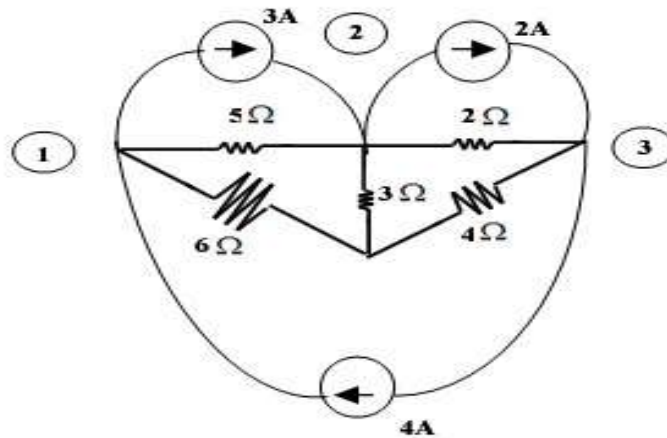
17. Calculate the equivalent resistance across AB



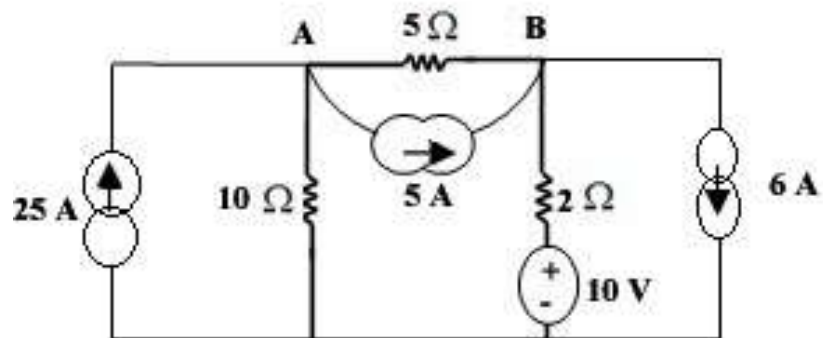
18. Calculate the current through the circuit using Mesh analysis?



19. Calculate the node voltage for the given circuit?



20. Calculate the voltage at the node A and B in the given circuit?

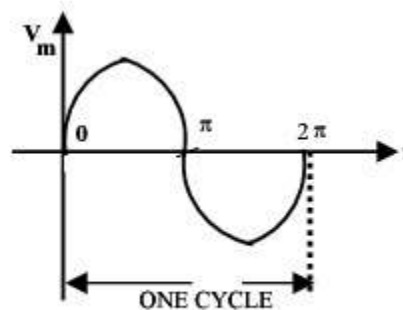


Unit-II AC CIRCUITS**PART A:**

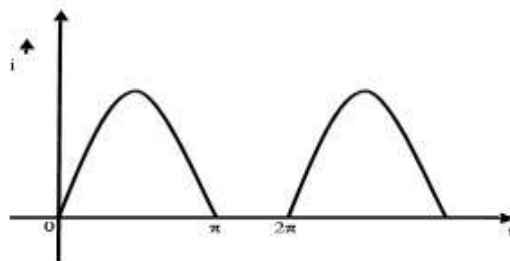
1. Write the expression for real and reactive power in a three phase system.
2. Define form factor.
3. Define average value.
4. Define root mean square value.
5. Define active power.
6. Define peak factor.
7. Define phase.
8. Define average value.
9. In a balanced Y-connected power system, calculate the phase voltage if the line voltage is 480V.
10. Draw the power triangle and explain the three types of power in a.c circuit.
11. List out the advantages of three phase over single phase power system.

PART B:

1. Calculate average value, RMS value, Peak factor and Form factor for the given Sinusoidal waveform.

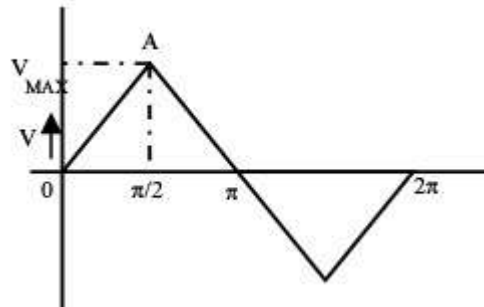


2. Derive the expression for R.M.S. value, Average value, Form factor, Peak factor of half wave rectified waveform is shown in below figure.

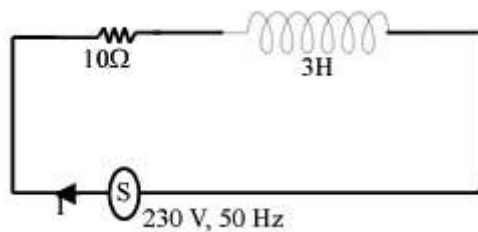


3. An alternating voltage of frequency 50 Hz has an R.M.S. value of 200 V. Find the instantaneous value of voltage after 2.5 milliseconds. Also determine the time for the instantaneous voltage of 150 V.

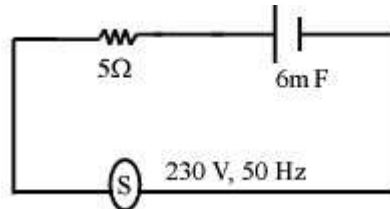
4. Determine the average value and the R.M.S. value of the triangle wave as show in the below figure.



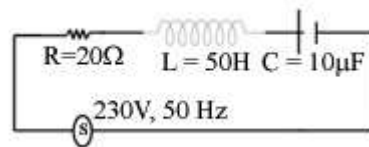
5. Calculate power factor, impedance, circuit and power of the given R_L circuit. The resistance is $10\ \Omega$, Inductance is $3H$ and voltage supply across the circuit



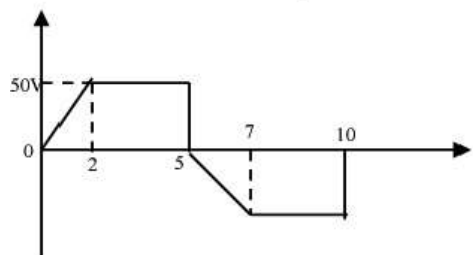
6. Calculate impedance z , power p and power factor for the following circuit



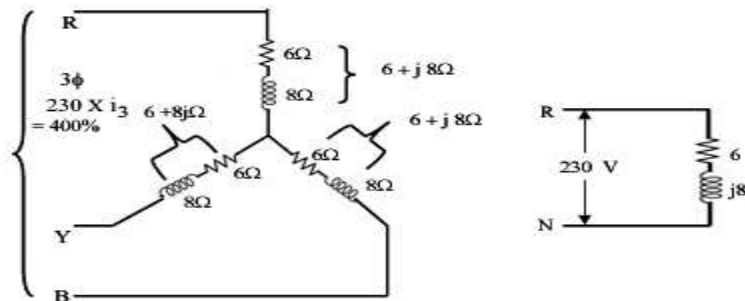
7. Calculate the impedance, current for the given RLC circuit $R = 20\ \Omega$, $L = 50\ H$, $C = 10\ \mu F$, $V = 230V$, $f = 50Hz$



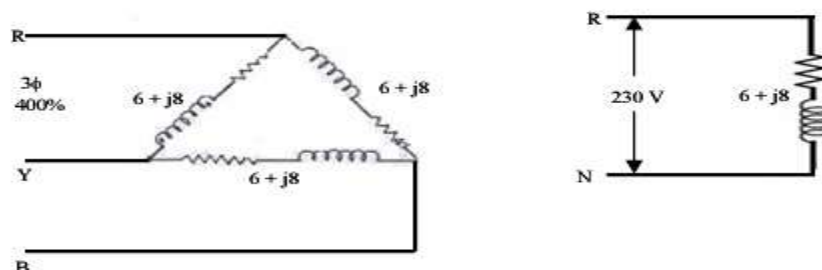
8. Find the average value of the voltage waveform shown in the figure



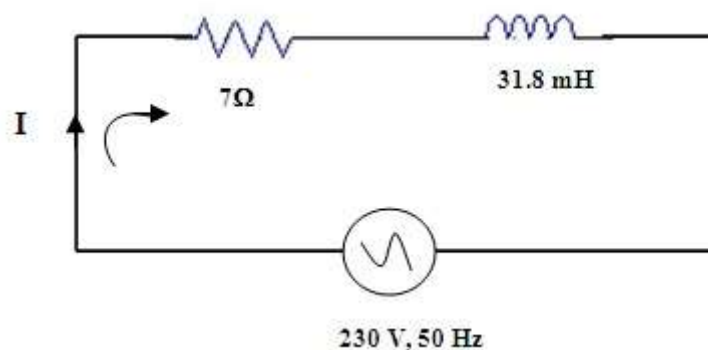
9. A balanced three phase star connected load consists of 6Ω resistor and 8Ω reactor (inductive) in each phase the supply is 230 V , 3ϕ , 50 Hz . Find (Part –II April 2010)
- The phase current
 - line current
 - total power



10. A balanced 3ϕ delta connected load consists of 6Ω resistor and 8Ω reactor (inductive) in each phase. The supply is 230 V , 3ϕ , 50 Hz , Find a) phase current, b) line current, c) power.



11. Find the power measured by two wattmeter method.
12. A coil having a resistance of 7Ω and inductance of 31.8 mH is connected to 230 V , 50 Hz supply. Calculate the circuit current power factor and power consumed.



Unit-III ELECTRICAL MACHINES & POWER PLANTS**PART A:**

1. List out the different types of DC motors.
2. What are the various sources of electrical energy?
3. What is a distribution system?
4. What is a DC generator?
5. State Faraday's laws of electromagnetic induction.
6. State Fleming's right hand rule.
7. What is the main function of circuit breakers?
8. What are the essential components of DC machine?
9. What is an ideal transformer?
10. What is the use of fuse?
11. What are the types of synchronous generator?
12. Mention the uses of circuit breakers?
13. State Fleming's right and left hand rule.
14. How does DC generator work?

PART B:

1. Explain the construction, working principle of any one type of single phase induction motor and deduce the emf equation?
2. Explain the construction, working principle of transformer?
3. Explain with block diagram of thermal power plant?
4. Explain the principle of operation of DC generator?
5. With block diagram explain the hydraulic plant.
6. Enumerate the difference between synchronous generator and induction generator. Explain in detail about the construction, working principle of synchronous generator and name few applications of it.